

NUCLEAR ATTACK SUBMARINE

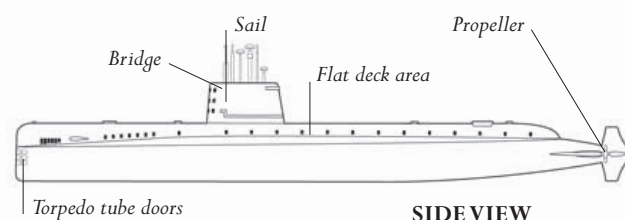
USS NAUTILUS

When the *Nautilus* entered service in 1954, it was the world's first nuclear-powered submarine. Its advent in the early part of the Cold War (see pp.384–85) marked a revolutionary step in naval warfare.

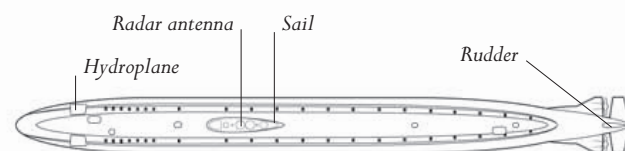
Three reactors enabled the *Nautilus* to travel for long periods without having to surface or refuel, unlike conventional submarines. In 1955, the ship set a new record for a submerged voyage, covering 2,222km (1,381 miles) in 90 hours. Three years later it became the first submarine to travel under the ice cap to the North Pole.

On the *Nautilus*'s upper deck was the attack centre – the heart of the ship's role as a warship. Forward of that were captain's and officers' quarters. The lower deck housed the control room, from where the ship was steered when under water, and the crew's mess and galley. The engines and reactors were near the stern.

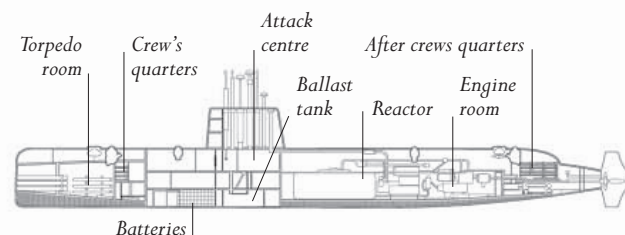
The *Nautilus* was not especially fast under water and carried only conventional torpedoes. Apart from the reactors, its major innovations were dispensing with a deck gun, providing a bunk for each crew member, and (on its 1958 sub-Arctic trip) an inertial navigation system. By 1959, the *Nautilus* was being superseded by submarines that, through improved hull design and the use of new materials, could travel faster and reach greater depths than before. Nuclear submarines would soon dominate naval warfare, either as torpedo-carrying attack submarines or as nuclear missile carriers. Retired in 1980, the *Nautilus* is now a museum.



SIDE VIEW



PLAN VIEW

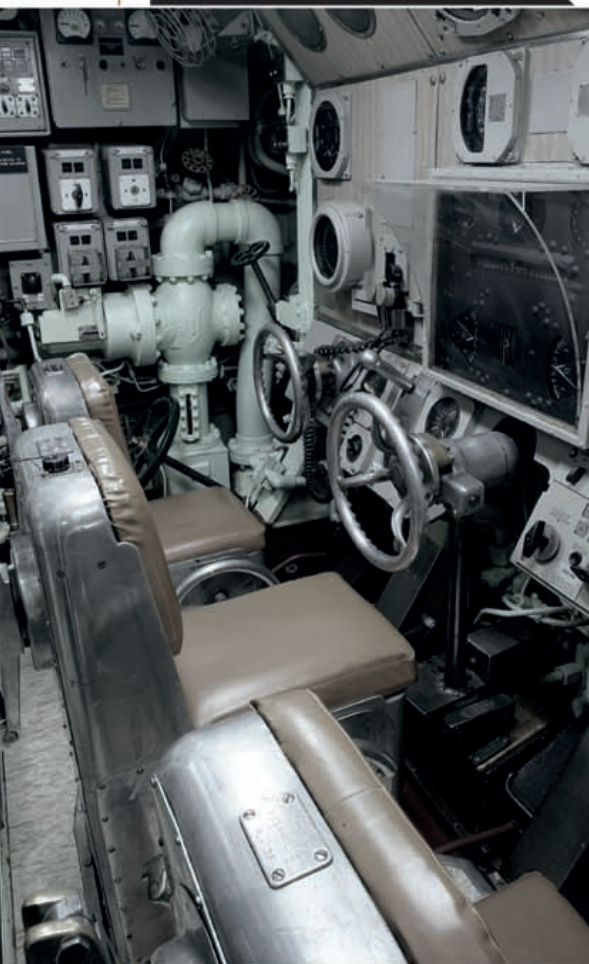


CROSS-SECTION

▲ USS NAUTILUS

Displacing 3,533 tons (4,092 tons submerged), the *Nautilus* was 98.6m (323½ft) long and carried a crew of 116. It had a top speed of 23 knots and could dive to a maximum depth of 213m (700ft).

OPERATING SYSTEMS



▲ CONTROL ROOM

Two planesmen operated the hydroplanes to adjust the ship's angle and depth. To their right, the helmsman operated the rudder.



▲ ALARMS

Alerts were sounded before the craft began to dive or if there was danger of a collision.



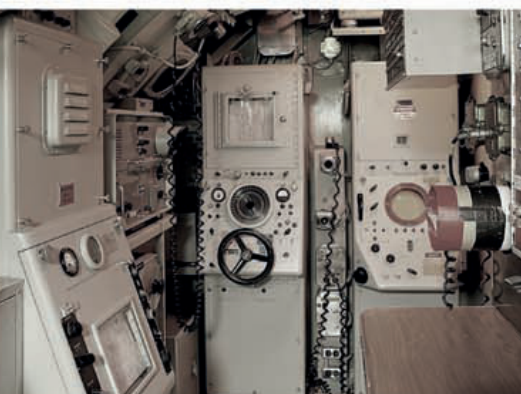
▲ WATER LEVELS

Dials indicated the amount of water contained in the ship's ballast tanks.



▲ DISPLAY PANEL

Prior to diving, this panel was checked to determine the status of the craft's numerous hatches and vents.



▲ SONAR ROOM

Sonar equipment and displays were used to locate ships and other submarines, follow target objects, and avoid underwater collisions.

► SUBMARINE SAIL

When travelling on the surface, the *Nautilus* was steered from the bridge, located at the top of the submarine's "sail" (conning tower).

▲ ELECTRONIC COUNTER-MEASURES

These devices detected signals from other vessels and helped foil attempts to track the ship from its own transmissions.

